



Configuration des équipements

1. Configuration DNS :

The screenshot shows a web-based configuration interface for a DNS server. The window title is "DNS SERVER 192.168.50.100 2501:0:CAFE::B". The interface has several tabs: "Physical", "Config", "Services", "Desktop", "Programming", and "Attributes". The "Services" tab is active, and within it, the "DNS" service is selected. On the left, a "SERVICES" sidebar lists various services like HTTP, DHCP, and DNS, with "DNS" highlighted. The main area shows the "DNS" configuration. The "DNS Service" is turned "On". Under "Resource Records", there is a form to add records with fields for "Name", "Type" (set to "AAAA Record"), and "Address". Below this is a table of existing records. At the bottom, there is a "DNS Cache" button and a "Top" link.

No.	Name	Type	Detail
0	cyberconseil.com	A Record	132.186.32.34
1	cyberconseil.com	A Record	132.186.32.49
2	cyberconseil.com	A Record	132.186.32.37
3	cyberconseil.com	AAAA Record	2901:0:CAFE::B

2. Configuration des routeurs (comprenant les tables de routage, les sous-interfaces et les NAT) :

Routeur Accueil :

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Router

!

!

!

!

!

!

!

!

no ip cef

ipv6 unicast-routing

!

no ipv6 cef

!

!

!

!

license udi pid CISCO2911/K9 sn FTX15245673-

!

!

!

!

!

!

```
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
  no ip address  
  duplex auto  
  speed auto  
  shutdown  
!  
interface GigabitEthernet0/1  
  no ip address  
  duplex auto  
  speed auto  
  shutdown  
!  
interface GigabitEthernet0/2  
  no ip address  
  duplex auto  
  speed auto  
  shutdown  
!  
interface Serial0/0/0
```

ip address 132.186.32.33 255.255.255.252

ipv6 address 2001:0:CAFE:1::/127

ipv6 enable

ipv6 ospf 1 area 1

!

interface Serial0/0/1

no ip address

clock rate 2000000

shutdown

!

interface Serial0/1/0

no ip address

clock rate 2000000

shutdown

!

interface Serial0/1/1

no ip address

clock rate 2000000

shutdown

!

interface Serial0/2/0

no ip address

clock rate 2000000

shutdown

!

interface Serial0/2/1

no ip address

clock rate 2000000

shutdown

!

interface Serial0/3/0

ip address 132.186.32.46 255.255.255.252

ipv6 address 2001:0:CAFE:2::/127

ipv6 enable

ipv6 ospf 1 area 1

clock rate 2000000

!

interface Serial0/3/1

no ip address

clock rate 2000000

shutdown

!

interface Vlan1

no ip address

shutdown

!

router ospf 1

router-id 1.1.1.1

log-adjacency-changes

network 132.186.32.32 0.0.0.3 area 1

network 132.186.32.44 0.0.0.3 area 1

!

ipv6 router ospf 1

router-id 1.1.1.1

log-adjacency-changes

!

ip classless

!

ip flow-export version 9

!

!

!

```
!  
!  
!  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!  
end
```

Routeur Local Technique :

```
!  
version 15.1  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Router  
!  
!  
!  
!  
!
```

```
!  
!  
!  
no ip cef  
ipv6 unicast-routing  
!  
no ipv6 cef  
!  
!  
!  
!  
license udi pid CISCO2911/K9 sn FTX15244FBE-  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0
```

no ip address

duplex auto

speed auto

shutdown

!

interface GigabitEthernet0/1

no ip address

duplex auto

speed auto

shutdown

!

interface GigabitEthernet0/2

no ip address

duplex auto

speed auto

shutdown

!

interface Serial0/0/0

no ip address

clock rate 2000000

shutdown

!

interface Serial0/0/1

ip address 132.186.32.50 255.255.255.252

ipv6 address 2001:0:CAFE:5::1/127

ipv6 enable

ipv6 ospf 1 area 1

clock rate 2000000

!

interface Serial0/1/0

no ip address

clock rate 2000000

shutdown

!

interface Serial0/1/1

no ip address

clock rate 2000000

shutdown

!

interface Serial0/2/0

ip address 132.186.32.42 255.255.255.252

ipv6 address 2001:0:CAFE:4::1/127

ipv6 enable

ipv6 ospf 1 area 1

clock rate 2000000

!

interface Serial0/2/1

no ip address

clock rate 2000000

shutdown

!

interface Serial0/3/0

ip address 132.186.32.45 255.255.255.252

ipv6 address 2001:0:CAFE:2::1/127

ipv6 enable

ipv6 ospf 1 area 1

!

interface Serial0/3/1

no ip address

clock rate 2000000

shutdown

!

```
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 4.4.4.4
log-adjacency-changes
network 132.186.32.40 0.0.0.3 area 1
network 132.186.32.44 0.0.0.3 area 1
network 132.186.32.48 0.0.0.3 area 1
!
ipv6 router ospf 1
router-id 4.4.4.4
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
```

```
!  
!  
!  
end
```

Routeur Salle Serveur :

```
!  
version 15.1  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Router  
!  
!  
!  
!  
!  
!  
!  
!  
no ip cef  
ipv6 unicast-routing  
!  
no ipv6 cef  
!  
!  
!
```

```
!  
license udi pid CISCO2911/K9 sn FTX1524163K-
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
spanning-tree mode pvst
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
interface GigabitEthernet0/0
```

```
no ip address
```

```
duplex auto
```

```
speed auto
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.0.0.1 255.0.0.0
```

```
ip nat inside
```

```
duplex auto
```

```
speed auto
```

ipv6 address 2901:0:CAFE::/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface GigabitEthernet0/2

no ip address

duplex auto

speed auto

shutdown

!

interface Serial0/0/0

ip address 132.186.32.34 255.255.255.252

ip nat outside

ipv6 address 2001:0:CAFE:1::1/127

ipv6 enable

ipv6 ospf 1 area 1

clock rate 2000000

!

interface Serial0/0/1

ip address 132.186.32.49 255.255.255.252

ip nat outside

ipv6 address 2001:0:CAFE:5::/127

ipv6 enable

ipv6 ospf 1 area 1

!

interface Serial0/1/0

ip address 132.186.32.37 255.255.255.252

ip nat outside

ipv6 address 2001:0:CAFE:3::/127

ipv6 enable

ipv6 ospf 1 area 1

```
!  
interface Serial0/1/1  
no ip address  
clock rate 2000000  
shutdown
```

```
!  
interface Serial0/2/0  
no ip address  
clock rate 2000000  
shutdown
```

```
!  
interface Serial0/2/1  
no ip address  
clock rate 2000000  
shutdown
```

```
!  
interface Serial0/3/0  
no ip address  
clock rate 2000000  
shutdown
```

```
!  
interface Serial0/3/1  
no ip address  
clock rate 2000000  
shutdown
```

```
!  
interface Vlan1  
no ip address  
shutdown
```

```
!  
router ospf 1
```

```
router-id 2.2.2.2
log-adjacency-changes
network 132.186.32.32 0.0.0.3 area 1
network 132.186.32.48 0.0.0.3 area 1
network 132.186.32.36 0.0.0.3 area 1
network 10.0.0.0 0.255.255.255 area 1
!
ipv6 router ospf 1
router-id 2.2.2.2
log-adjacency-changes
!
ip nat inside source list 1 interface GigabitEthernet0/1 overload
ip nat inside source static tcp 10.0.0.2 80 132.186.32.37 80
ip nat inside source static tcp 10.0.0.2 443 132.186.32.37 443
ip nat inside source static tcp 10.0.0.2 80 132.186.32.49 80
ip nat inside source static tcp 10.0.0.2 443 132.186.32.49 443
ip nat inside source static tcp 10.0.0.2 80 132.186.32.34 80
ip nat inside source static tcp 10.0.0.2 443 132.186.32.34 443
ip classless
!
ip flow-export version 9
!
!
access-list 1 permit 10.0.0.0 0.255.255.255
!
!
!
!
!
line con 0
!
```

```
line aux 0
!
line vty 0 4
login
!
!
!
end
```

Routeur Bureaux :

```
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
```


!

!

!

license udi pid CISCO2911/K9 sn FTX15244SBY-

!

!

!

!

!

!

!

!

!

!

!

spanning-tree mode pvst

!

!

!

!

!

!

interface GigabitEthernet0/0

no ip address

duplex auto

speed auto

shutdown

!

interface GigabitEthernet0/1

no ip address

duplex auto

speed auto

shutdown

!

interface GigabitEthernet0/2

no ip address

duplex auto

speed auto

shutdown

!

interface GigabitEthernet0/0/0

ip address 172.16.10.9 255.255.255.252

ip nat inside

ipv6 address 2001:0:CAFE:6::/127

ipv6 enable

ipv6 ospf 1 area 1

!

interface Serial0/1/0

ip address 132.186.32.38 255.255.255.252

ip nat outside

ipv6 address 2001:0:CAFE:3::1/127

ipv6 enable

ipv6 ospf 1 area 1

clock rate 2000000

!

interface Serial0/1/1

no ip address

clock rate 2000000

shutdown

!

interface Serial0/2/0

ip address 132.186.32.41 255.255.255.252

```
ipv6 address 2001:0:CAFE:4::/127
ipv6 enable
ipv6 ospf 1 area 1
!
interface Serial0/2/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 3.3.3.3
log-adjacency-changes
network 132.186.32.36 0.0.0.3 area 1
network 132.186.32.40 0.0.0.3 area 1
network 172.16.10.8 0.0.0.3 area 1
!
ipv6 router ospf 1
```

```
router-id 3.3.3.3
log-adjacency-changes
!
ip nat inside source list 1 interface Serial0/1/0 overload
ip classless
!
ip flow-export version 9
!
!
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255
access-list 1 permit 192.168.50.0 0.0.0.255
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

3. Configuration des switchs (comprenant les paramètres DHCP, les VLAN, le LACP et les ACL) :

3.1 Configuration des switchs de type layer 3 :

Switch Core :

!

version 16.3.2

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Switch

!

!

!

ip dhcp excluded-address 192.168.10.1

ip dhcp excluded-address 192.168.20.1

ip dhcp excluded-address 192.168.30.1

ip dhcp excluded-address 192.168.40.1

!

ip dhcp pool direction

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 192.168.50.100

ip dhcp pool rh

network 192.168.20.0 255.255.255.0

default-router 192.168.20.1

dns-server 192.168.50.100

ip dhcp pool commerciaux

network 192.168.30.0 255.255.255.0

[illegible]

!

!

!

!

!

interface GigabitEthernet1/0/1

!

interface GigabitEthernet1/0/2

!

interface GigabitEthernet1/0/3

!

interface GigabitEthernet1/0/4

!

interface GigabitEthernet1/0/5

!

interface GigabitEthernet1/0/6

!

interface GigabitEthernet1/0/7

!

interface GigabitEthernet1/0/8

!

interface GigabitEthernet1/0/9

!

interface GigabitEthernet1/0/10

!

interface GigabitEthernet1/0/11

!

interface GigabitEthernet1/0/12

!

interface GigabitEthernet1/0/13

!

```
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
no switchport
ip address 172.16.10.1 255.255.255.252
ipv6 address 2001:0:CAFE:7::/127
ipv6 enable
ipv6 ospf 1 area 1
!
interface GigabitEthernet1/1/2
no switchport
```



```
ip address 172.16.10.5 255.255.255.252
ipv6 address 2001:0:CAFE:8::/127
ipv6 enable
ipv6 ospf 1 area 1
!
interface GigabitEthernet1/1/3
no switchport
ip address 172.16.10.10 255.255.255.252
ipv6 address 2001:0:CAFE:6::1/127
ipv6 enable
ipv6 ospf 1 area 1
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 5.5.5.5
log-adjacency-changes
network 192.168.10.0 0.0.0.255 area 1
network 192.168.20.0 0.0.0.255 area 1
network 192.168.30.0 0.0.0.255 area 1
network 192.168.40.0 0.0.0.255 area 1
network 192.168.50.0 0.0.0.255 area 1
network 172.16.10.8 0.0.0.3 area 1
network 172.16.10.4 0.0.0.3 area 1
network 172.16.10.0 0.0.0.3 area 1
!
ipv6 router ospf 1
```

```
router-id 5.5.5.5
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
!
end
```

Switch Distributed 1 :

```
!
version 16.3.2
```

```
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
!
```

```
hostname Switch
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
no ip cef
```

```
ip routing
```

```
!
```

```
ipv6 unicast-routing
```

```
!
```

```
no ipv6 cef
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

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!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
!  
interface Port-channel1  
    switchport mode trunk  
!  
interface GigabitEthernet1/0/1  
    switchport mode trunk  
!  
interface GigabitEthernet1/0/2  
    switchport mode trunk  
!  
interface GigabitEthernet1/0/3  
    switchport mode trunk  
!  
interface GigabitEthernet1/0/4  
    switchport mode trunk  
!  
interface GigabitEthernet1/0/5  
    switchport mode trunk  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8
```

```
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
    switchport mode trunk  
    channel-protocol lacp  
    channel-group 1 mode active  
!  
interface GigabitEthernet1/0/22
```

```
switchport mode trunk
channel-protocol lacp
channel-group 1 mode active
!
interface GigabitEthernet1/0/23
switchport mode trunk
channel-protocol lacp
channel-group 1 mode active
!
interface GigabitEthernet1/0/24
switchport mode trunk
channel-protocol lacp
channel-group 1 mode active
!
interface GigabitEthernet1/1/1
no switchport
ip address 172.16.10.2 255.255.255.252
ip ospf priority 255
ipv6 address 2001:0:CAFE:7::1/127
ipv6 enable
ipv6 ospf 1 area 1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
```

!

interface Vlan10

mac-address 0002.167e.ed01

ip address 192.168.10.1 255.255.255.0

ip helper-address 172.16.10.1

ipv6 address FE80::110:1 link-local

ipv6 address 2101:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface Vlan20

mac-address 0002.167e.ed02

ip address 192.168.20.1 255.255.255.0

ip helper-address 172.16.10.1

ipv6 address FE80::120:1 link-local

ipv6 address 2201:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface Vlan30

mac-address 0002.167e.ed03

ip address 192.168.30.1 255.255.255.0

ip helper-address 172.16.10.1

ipv6 address FE80::130:1 link-local

ipv6 address 2301:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface Vlan40

mac-address 0002.167e.ed04

ip address 192.168.40.1 255.255.255.0

```
ip helper-address 172.16.10.1
ip access-group 100 out
ipv6 address FE80::140:1 link-local
ipv6 address 2401:0:CAFE::A/64
ipv6 enable
ipv6 ospf 1 area 1
!
interface Vlan50
mac-address 0002.167e.ed05
ip address 192.168.50.1 255.255.255.0
ip helper-address 172.16.10.1
ipv6 address FE80::150:1 link-local
ipv6 address 2501:0:CAFE::A/64
ipv6 enable
ipv6 ospf 1 area 1
!
interface Vlan110
mac-address 0002.167e.ed06
no ip address
!
router ospf 1
router-id 6.6.6.6
log-adjacency-changes
network 172.16.10.0 0.0.0.3 area 1
network 192.168.10.0 0.0.0.255 area 1
network 192.168.20.0 0.0.0.255 area 1
network 192.168.30.0 0.0.0.255 area 1
network 192.168.40.0 0.0.0.255 area 1
network 192.168.50.0 0.0.0.255 area 1
!
ipv6 router ospf 1
```



```
router-id 6.6.6.6
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
access-list 100 permit tcp 192.168.40.0 0.0.0.255 any established
access-list 100 permit udp 192.168.40.0 0.0.0.255 host 192.168.50.100 eq domain
access-list 100 permit udp host 192.168.50.100 eq domain 192.168.40.0 0.0.0.255
access-list 100 permit icmp any 192.168.40.0 0.0.0.255 echo-reply
access-list 100 permit tcp any any established
access-list 100 permit tcp 192.168.40.0 0.0.0.255 any
access-list 100 deny ip any any
ipv6 access-list ipv6acl
permit tcp 2401:0:CAFE::/64 any
permit udp 2401:0:CAFE::/64 host 2501:0:CAFE::B eq domain
permit udp host 2501:0:CAFE::B eq domain 2401:0:CAFE::/64 eq domain
permit icmp any 2401:0:CAFE::/64 echo-reply
permit tcp any any
deny ipv6 any any
!
!
!
!
!
line con 0
!
line aux 0
!
```

```
line vty 0 4
```

```
login
```

```
!
```

```
!
```

```
!
```

```
!
```

```
end
```

```
Switch Distributed 2 :
```

```
!
```

```
version 16.3.2
```

```
no service timestamps log datetime msec
```

```
no service timestamps debug datetime msec
```

```
no service password-encryption
```

```
!
```

```
hostname Switch
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
no ip cef
```

```
ip routing
```

```
!
```

```
ipv6 unicast-routing
```

```
!
```

```
no ipv6 cef
```

!
!
!
!
!
!
!
!
!
!
!
!
!
!

spanning-tree mode pvst

!
!
!
!
!
!
!

interface Port-channel1

switchport mode trunk

!

interface GigabitEthernet1/0/1

switchport mode trunk

!

interface GigabitEthernet1/0/2

switchport mode trunk

!

interface GigabitEthernet1/0/3

```
switchport mode trunk
!
interface GigabitEthernet1/0/4
switchport mode trunk
!
interface GigabitEthernet1/0/5
switchport mode trunk
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
```

```
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
    switchport mode trunk  
    channel-protocol lacp  
    channel-group 1 mode active  
!  
interface GigabitEthernet1/0/22  
    switchport mode trunk  
    channel-protocol lacp  
    channel-group 1 mode active  
!  
interface GigabitEthernet1/0/23  
    switchport mode trunk  
    channel-protocol lacp  
    channel-group 1 mode active  
!  
interface GigabitEthernet1/0/24  
    switchport mode trunk  
    channel-protocol lacp  
    channel-group 1 mode active  
!  
interface GigabitEthernet1/1/1  
!  
interface GigabitEthernet1/1/2  
    no switchport
```

ip address 172.16.10.6 255.255.255.252

ip ospf priority 255

ipv6 address 2001:0:CAFE:8::1/127

ipv6 enable

ipv6 ospf 1 area 1

!

interface GigabitEthernet1/1/3

!

interface GigabitEthernet1/1/4

!

interface Vlan1

no ip address

shutdown

!

interface Vlan10

mac-address 0050.0f8a.8501

ip address 192.168.10.1 255.255.255.0

ip helper-address 172.16.10.5

ipv6 address FE80::110:1 link-local

ipv6 address 2101:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface Vlan20

mac-address 0050.0f8a.8502

ip address 192.168.20.1 255.255.255.0

ip helper-address 172.16.10.5

ipv6 address FE80::120:1 link-local

ipv6 address 2201:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface Vlan30

mac-address 0050.0f8a.8503

ip address 192.168.30.1 255.255.255.0

ip helper-address 172.16.10.5

ipv6 address FE80::130:1 link-local

ipv6 address 2301:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface Vlan40

mac-address 0050.0f8a.8504

ip address 192.168.40.1 255.255.255.0

ip helper-address 172.16.10.5

ip access-group 100 out

ipv6 address FE80::140:1 link-local

ipv6 address 2401:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

interface Vlan50

mac-address 0050.0f8a.8505

ip address 192.168.50.1 255.255.255.0

ip helper-address 172.16.10.5

ipv6 address FE80::150:1 link-local

ipv6 address 2501:0:CAFE::A/64

ipv6 enable

ipv6 ospf 1 area 1

!

router ospf 1

router-id 7.7.7.7

```
log-adjacency-changes
network 172.16.10.4 0.0.0.3 area 1
network 192.168.10.0 0.0.0.255 area 1
network 192.168.20.0 0.0.0.255 area 1
network 192.168.30.0 0.0.0.255 area 1
network 192.168.40.0 0.0.0.255 area 1
network 192.168.50.0 0.0.0.255 area 1
!
ipv6 router ospf 1
router-id 7.7.7.7
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
access-list 100 permit icmp any 192.168.40.0 0.0.0.255 echo-reply
access-list 100 permit tcp 192.168.40.0 0.0.0.255 any established
access-list 100 permit udp 192.168.40.0 0.0.0.255 host 192.168.50.100 eq domain
access-list 100 permit udp host 192.168.50.100 eq domain 192.168.40.0 0.0.0.255
access-list 100 permit tcp any any established
access-list 100 permit tcp 192.168.40.0 0.0.0.255 any
access-list 100 deny ip any any
ipv6 access-list ipv6acl
permit tcp 2401:0:CAFE::/64 any
permit udp 2401:0:CAFE::/64 host 2501:0:CAFE::B eq domain
permit udp host 2501:0:CAFE::B eq domain 2401:0:CAFE::/64 eq domain
permit icmp any 2401:0:CAFE::/64 echo-reply
permit tcp any any
deny ipv6 any any
```



```
!  
!  
!  
!  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!  
!  
end
```

3.2 Configuration des switchs de type layer 2 :

Switch Salle Serveur :

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Switch  
!  
!  
!
```

```
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
!  
interface FastEthernet0/2  
!  
interface FastEthernet0/3  
!  
interface FastEthernet0/4  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
!  
interface FastEthernet0/11  
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13
```

```
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
!  
interface GigabitEthernet0/2  
!  
interface Vlan1  
no ip address  
shutdown  
!
```

```
!  
!  
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
!  
!  
!  
end
```

Switch Direction :

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Switch  
!  
!  
!  
!  
!  
!
```

```
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport access vlan 10
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
```

```
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
!  
interface GigabitEthernet0/2  
!  
interface Vlan1  
no ip address  
shutdown  
!  
!  
!
```

```
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
!  
!  
!  
end
```

Switch RH :

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Switch  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id
```

```
!  
interface FastEthernet0/1  
  switchport access vlan 20  
!  
interface FastEthernet0/2  
!  
interface FastEthernet0/3  
!  
interface FastEthernet0/4  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
!  
interface FastEthernet0/11  
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15
```



```
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
!  
interface GigabitEthernet0/2  
!  
interface Vlan1  
no ip address  
shutdown  
!  
!  
!  
!  
line con 0
```

```
!  
line vty 0 4  
  login  
line vty 5 15  
  login  
!  
!  
!  
!  
end
```

Switch Commerciaux :

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Switch  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface FastEthernet0/1
```

```
switchport access vlan 30
```

```
!
```

```
interface FastEthernet0/2
```

```
!
```

```
interface FastEthernet0/3
```

```
!
```

```
interface FastEthernet0/4
```

```
!
```

```
interface FastEthernet0/5
```

```
!
```

```
interface FastEthernet0/6
```

```
!
```

```
interface FastEthernet0/7
```

```
!
```

```
interface FastEthernet0/8
```

```
!
```

```
interface FastEthernet0/9
```

```
!
```

```
interface FastEthernet0/10
```

```
!
```

```
interface FastEthernet0/11
```

```
!
```

```
interface FastEthernet0/12
```

```
!
```

```
interface FastEthernet0/13
```

```
!
```

```
interface FastEthernet0/14
```

```
!
```

```
interface FastEthernet0/15
```

```
!
```

```
interface FastEthernet0/16
```

```
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
!  
interface GigabitEthernet0/2  
!  
interface Vlan1  
no ip address  
shutdown  
!  
!  
!  
!  
line con 0  
!  
line vty 0 4
```

```
login
line vty 5 15
login
!
!
!
!
end
```

Switch IT/Cybersécurité :

```
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 40
```

```
!  
interface FastEthernet0/2  
!  
interface FastEthernet0/3  
!  
interface FastEthernet0/4  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
!  
interface FastEthernet0/11  
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!
```

```
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
!
!
!
line con 0
!
line vty 0 4
login
```

```
line vty 5 15
```

```
login
```

```
!
```

```
!
```

```
!
```

```
!
```

```
end
```

Switch du serveur DNS :

```
!
```

```
version 15.0
```

```
no service timestamps log datetime msec
```

```
no service timestamps debug datetime msec
```

```
no service password-encryption
```

```
!
```

```
hostname Switch
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
spanning-tree mode pvst
```

```
spanning-tree extend system-id
```

```
!
```

```
interface FastEthernet0/1
```

```
switchport access vlan 50
```

```
!
```



```
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
```

```
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
!  
interface GigabitEthernet0/2  
!  
interface Vlan1  
no ip address  
shutdown  
!  
!  
!  
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15
```

login

!

!

!

!

end